Given (for all proofs below)

$V = IR + E_{\text{mf}}$ by Ohm's Law with back EMF for a DC motor

$E_{\text{mf}} = \omega\kappa_e$ by the Back EMF equation of a motor

$\tau = I\kappa_t$ by the Motor torque equation

$\tau = J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)$ by the Rotational equation of motion

The value of ω and α (Only for the proof directly below this)

Find a function V_{ff}, voltage, for a feedforward controller

$V_{\text{ff}} \coloneqq V$ for this proof only for notational ease of use

$\omega \coloneqq \omega_{\text{ref}}$ for this proof only for notational ease of use

$\alpha \coloneqq \alpha_{\text{ref}}$ for this proof only for notational ease of use

$V = IR + E_{\text{mf}}$, $E_{\text{mf}} = \omega\kappa_e$, $\tau = I\kappa_t$, $\tau = J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)$, and the value of ω and α because they are given

$I\kappa_t = J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)$ By substitution

$I = \frac{J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)}{\kappa_t}$ By Division

$-IR = E_{\text{mf}} - V$ By Subtraction

$I = \frac{E_{\text{mf}} - V}{-R}$ By Division

$\frac{E_{\text{mf}} - V}{-R} = \frac{J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)}{\kappa_t}$ By substitution

$\frac{\omega\kappa_e - V}{-R} = \frac{J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)}{\kappa_t}$ By substitution

$\omega\kappa_e - V = \frac{J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)}{\kappa_t}(-R)$ By multiplication

$-V = \frac{J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)}{\kappa_t}(-R) - \omega\kappa_e$ By subtraction

$V = -\left(\frac{J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)}{\kappa_t}(-R) - \omega\kappa_e\right)$ By division

$V = \frac{J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)}{\kappa_t}(R) + \omega\kappa_e$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{B\omega R}{\kappa_t} + \frac{\tau_{\text{load}}\ \text{sign}(\omega) R}{\kappa_t} + \frac{\omega\kappa_e\kappa_t}{\kappa_t}$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{B\omega R + \omega\kappa_e\kappa_t}{\kappa_t} + \frac{\tau_{\text{load}}\ \text{sign}(\omega) R}{\kappa_t}$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{\omega(BR + \kappa_e\kappa_t)}{\kappa_t} + \frac{\tau_{\text{load}}\ \text{sign}(\omega) R}{\kappa_t}$ By substitution

Define K_v by $K_v = \frac{BR + \kappa_e\kappa_t}{\kappa_t}$

Define K_a by $K_a = \frac{JR}{\kappa_t}$

Define K_s by $K_s = \frac{\tau_{\text{load}} R}{\kappa_t}$

$V = K_a\alpha + K_v\omega + K_s\ \text{sign}(\omega)$ by substitution

$V_{\text{ff}} = K_a\alpha_{\text{ref}} + K_v\omega_{\text{ref}} + K_s\ \text{sign}(\omega_{\text{ref}})$ by substitution

Define a function V_{ff} by $V_{\text{ff}}(\omega_{\text{ref}}, \alpha_{\text{ref}}) = K_a\alpha_{\text{ref}} + K_v\omega_{\text{ref}} + K_s\ \text{sign}(\omega_{\text{ref}})$

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Given (Only for the proof directly below this)

The value of ω

Find the maximum α angular acceleration at some angular velocity ω

$V = IR + E_{\text{mf}}$, $E_{\text{mf}} = \omega\kappa_e$, $\tau = I\kappa_t$, $\tau = J\alpha + B\omega + \tau_{\text{load}}\ \text{sign}(\omega)$, and the value of ω because they are given

Note: The VEX motor controller has current limits so we must also account for those.

Define I_{min} and I_{max} as the current limits and define I_{clamp} as the clamped value of I

Define V_{max} as the maximum voltage and set it to V when finding the max acceleration

Define α_{max} as the maximum acceleration and set it to α when finding the max acceleration

$I_{\text{clamp}} = \min(\max(I, I_{\text{min}}), I_{\text{max}})$ by the definition of clamping

$I_{\text{clamp}} = \min\Big(\max\Big(\Big(\frac{E_{\text{mf}} - V_{\text{max}}}{-R}\Big), I_{\text{min}}\Big), I_{\text{max}}\Big)$ by substitution

$I_{\text{clamp}} = \min\Big(\max\Big(\Big(\frac{\omega\kappa_e - V_{\text{max}}}{-R}\Big), I_{\text{min}}\Big), I_{\text{max}}\Big)$ by substitution

Define $V_{\text{max}} = V$ as the max voltage

$I_{\text{clamp}} = \min\Big(\max\Big(\Big(\frac{\omega\kappa_e - V_{\text{max}}}{-R}\Big), I_{\text{min}}\Big), I_{\text{max}}\Big)$ by substitution

Restrict I by $I_{\text{clamp}} = I$ by the note

$-J\alpha_{\text{max}} = B\omega + \tau_{\text{load}}\ \text{sign}(\omega) - \tau$ by Subtraction

$\alpha_{\text{max}} = \frac{B\omega + \tau_{\text{load}}\ \text{sign}(\omega) - \tau}{-J}$ by Division

$\alpha_{\text{max}} = \frac{B\omega + \tau_{\text{load}}\ \text{sign}(\omega) - I\kappa_t}{-J}$ by Substitution

$\alpha_{\text{max}} = \frac{B\omega + \tau_{\text{load}}\ \text{sign}(\omega) - I_{\text{clamp}}\kappa_t}{-J}$ by Substitution

$\alpha_{\text{max}} = \frac{I_{\text{clamp}}\kappa_t - B\omega - \tau_{\text{load}}\ \text{sign}(\omega)}{J}$ by Substitution

Define a function α by $\alpha_{\text{max}} = \frac{I_{\text{clamp}}\kappa_t - B\omega - \tau_{\text{load}}\ \text{sign}(\omega)}{J}$

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Find $\omega(t)$ given that the motor is at a Constant Voltage

$\alpha_{\text{max}}(\omega) = \frac{V_{\text{max}} - K_v\omega - \ \text{sgn}(\omega)K_s}{K_a}$

Let $s = \text{sgn}(\omega) \in \{1, -1\}$

$\frac{d\omega}{dt} = \frac{V_{\text{max}} - K_v\omega - sK_s}{K_a}$

$\frac{d\omega}{dt} + \frac{K_v}{K_a}\omega = \frac{V_{\text{max}} - sK_s}{K_a}$

$I(t) = e^{\int \frac{K_v}{K_a} dt}$

$I(t) = e^{\frac{K_v}{K_a}t}$

$\omega(t) = \frac{1}{I(t)}\Big(\int_0^t I(\tau)Qd\tau + I(0)\omega_0\Big)$

$\omega(t) = \frac{1}{e^{\frac{K_v}{K_a}t}}\Big(\int_0^t e^{\frac{K_v}{K_a}\tau}\frac{V_{\text{max}} - sK_s}{K_a}d\tau + \omega_0\Big)$

$\omega(t) = \frac{1}{e^{\frac{K_v}{K_a}t}}\Big(\frac{V_{\text{max}} - sK_s}{K_a}\frac{K_a}{K_v}\Big(e^{\frac{K_v}{K_a}t} - 1\Big) + \omega_0\Big)$

$\omega(t) = \frac{V_{\text{max}} - sK_s}{K_v}\frac{e^{\frac{K_v}{K_a}t} - 1}{e^{\frac{K_v}{K_a}t}} + \omega_0\Big(e^{-\frac{K_v}{K_a}t}\Big)$

$\omega(t) = \frac{V_{\text{max}} - sK_s}{K_v} + \Big(\omega_0 - \frac{V_{\text{max}} - sK_s}{K_v}\Big)e^{-\frac{K_v}{K_a}t}$

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Define the plant (solve for the derivative of the state x where V is the desired input for the plant)

$x \coloneqq \omega$

$V \coloneqq u$

$\dot{x} = \dot{\omega}$ by the derivative

$\dot{x} = \alpha$ by substitution

$\therefore$ Find $\dot{x}$

$V_{\text{ff}} = K_a \alpha_{\text{ref}} + K_v \omega_{\text{ref}} + K_s \operatorname{sign}(\omega_{\text{ref}})$ because it is given

$V_{\text{ff}} \coloneqq u$ as the voltage is the input

$\alpha_{\text{ref}} \coloneqq \alpha$

$\omega_{\text{ref}} \coloneqq \omega$

$u = K_a \dot{x} + K_v x + K_s \operatorname{sign}(x)$ by substitution

$K_a \dot{x} = u - K_v x - K_s \operatorname{sign}(x)$ by subtraction

$\dot{x} = \frac{u - K_v x - K_s \operatorname{sign}(x)}{K_a}$ by division

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Find a continuous state-space model in the form

$\dot{x} = \boldsymbol{A}x + \boldsymbol{B}u$

$y = \boldsymbol{C}x + \boldsymbol{D}u$

$V_{\text{ff}} = K_a \alpha_{\text{ref}} + K_v \omega_{\text{ref}} + K_s \operatorname{sign}(\omega_{\text{ref}})$ because it is given

$u \coloneqq V_{\text{fb}}$

$V \coloneqq V_{\text{ff}} + V_{\text{fb}}$

$V = V_{\text{ff}} + u$

$\dot{x} = \frac{V - K_v x - K_s \operatorname{sign}(x)}{K_a}$ because it is given and V is the desired input for the plant

$e \coloneqq x - x_{\text{ref}}$ for x in the feedforward

$\dot{e} = \dot{x} - \dot{x}_{\text{ref}}$

$\dot{e} + \dot{x} = \frac{V - K_v x - K_s \operatorname{sign}(x)}{K_a} + \dot{e}$ by addition

$\dot{x} - \dot{x}_{\text{ref}} + \dot{x} = \frac{V\text{ff} + u - K_v \omega - K_s \operatorname{sign}(\omega)}{K_a} + \dot{x} - \dot{x}_{\text{ref}}$ by substitution

$\dot{x} - \dot{x}_{\text{ref}} + \dot{x} = \frac{K_a \alpha_{\text{ref}} + K_v \omega_{\text{ref}} + K_s \operatorname{sign}(\omega_{\text{ref}}) + u - K_v \omega - K_s \operatorname{sign}(\omega)}{K_a} + \dot{x} - \dot{x}_{\text{ref}}$ by substitution

$\dot{x} - \dot{x}_{\text{ref}} = \frac{K_a \alpha_{\text{ref}} + K_v \omega_{\text{ref}} + K_s \operatorname{sign}(\omega_{\text{ref}}) + u - K_v \omega - K_s \operatorname{sign}(\omega)}{K_a} - \dot{x}_{\text{ref}}$ by subtraction

$\dot{e} = \frac{u - K_v(\omega - \omega_{\text{ref}}) - K_s(\operatorname{sign}(\omega) - \operatorname{sign}(\omega_{\text{ref}})) + K_a \alpha_{\text{ref}}}{K_a} - \dot{\omega}_{\text{ref}}$ by substitution

$\dot{e} = \frac{u - K_v(x - x_{\text{ref}}) - K_s(\operatorname{sign}(x) - \operatorname{sign}(x_{\text{ref}})) + K_a \alpha_{\text{ref}}}{K_a} - \alpha_{\text{ref}}$ by substitution

Assume $\operatorname{sign}(x) = \operatorname{sign}(x_{\text{ref}}) \therefore \operatorname{sign}(x) - \operatorname{sign}(x_{\text{ref}}) = 0$ as this can be accounted for by the ff

$\dot{e} = \frac{u - K_v e}{K_a}$ by substitution

$\dot{e} = -\frac{K_v}{K_a} e + \frac{1}{K_a} u$ by substitution

$x \coloneqq e \quad \boldsymbol{A} \coloneqq -\frac{K_v}{K_a} \quad \boldsymbol{B} \coloneqq \frac{1}{K_a}$

$y := e \Rightarrow \boldsymbol{C} = 1$ and $\boldsymbol{D} = 0$

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Given

$\boldsymbol{A}_d = e^{\boldsymbol{A}_c T}$

$\boldsymbol{B}_d = \int_0^T e^{\boldsymbol{A}_c \tau} d\tau \boldsymbol{B}_c$

$\boldsymbol{C}_d = \boldsymbol{C}_c$

$\boldsymbol{D}_d = \boldsymbol{D}_c$

$\boldsymbol{A_c} = -\frac{K_v}{K_a}$

$\boldsymbol{B_c} = \frac{1}{K_a}$

$\boldsymbol{C_c} = 1$

$\boldsymbol{D_c} = 0$

Find a discrete state-space model in the form

$x_{k+1} = \boldsymbol{A}x_k + \boldsymbol{B}u_k$

$y_k = \boldsymbol{C}x_k + \boldsymbol{D}u_k$

$\boldsymbol{A}_d = e^{-\frac{K_v}{K_a} T}$ by substitution

$\boldsymbol{B}_d = \frac{1}{K_a} \int_0^T e^{-\frac{K_v}{K_a} \tau} d\tau$

$\boldsymbol{B}_d = -\frac{1}{K_a} \frac{K_a}{K_v} e^{-\frac{K_v}{K_a} \tau} |_0^T$

$\boldsymbol{B}_d = \frac{1}{K_v} \left(1 - e^{-\frac{K_v}{K_a} T} \right)$

$\boldsymbol{C}_d = 1$ by substitution

$\boldsymbol{D}_d = 0$ by substitution

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Given

$u_k^* = \arg\min_{u_k} \sum_{k=0}^{\infty} \left(x_k^T \boldsymbol{Q} x_k + \boldsymbol{u}^T \boldsymbol{R} \boldsymbol{u}_k \right)$

subject to $x_{k+1} = \boldsymbol{A}x_k + \boldsymbol{B}\boldsymbol{u_k}$ Find $\boldsymbol{K}$ based on $\boldsymbol{Q}$ and $\boldsymbol{R}$ in LQR

$\boldsymbol{A}^T \boldsymbol{S} \boldsymbol{A} - \boldsymbol{S} - \boldsymbol{A}^T \boldsymbol{S} \boldsymbol{B} (\boldsymbol{R} + \boldsymbol{B}^T \boldsymbol{S} \boldsymbol{B})^{-1} \boldsymbol{B}^T \boldsymbol{S} \boldsymbol{A} + Q = 0$ by the discrete algebraic Riccati equation

$\boldsymbol{K} = (\boldsymbol{R} + \boldsymbol{B}^T \boldsymbol{S} \boldsymbol{B})^{-1} \boldsymbol{B}^T \boldsymbol{S} \boldsymbol{A}$ by the discrete LQR result solved by the discrete algebraic Riccati equation

$\boldsymbol{A} := e^{-\frac{K_v}{K_a} T} := a$

$\boldsymbol{B} := \frac{1}{K_v} \left(1 - e^{-\frac{K_v}{K_a} T} \right) := b$

$\boldsymbol{C} = 1$ by substitution

$\boldsymbol{D} = 0$ by substitution

$a^2 S - S - \frac{a^2 b^2 S^2}{R + b^2 S} + Q = 0$ by substitution

$a^2 SR + a^2 b^2 S^2 - SR - b^2 S^2 + QR + Qb^2 S - a^2 b^2 S^2 = 0$ by multiplication

$a^2SR - SR - b^2S^2 + QR + Qb^2S = 0$ by substitution

$b^2S^2 + (R(1-a^2) - Qb^2)S - QR = 0$ by substitution

$Q, R, b^2 > 0 \therefore 4b^2QR > 0 \therefore \sqrt{(R(1-a^2) - Qb^2)^2 + 4b^2QR} > |R(1-a^2) - Qb^2| \therefore$ the smaller root is negative which is not possible as $S > 0$ so only the possitive root is valid

$S = \frac{Qb^2 - R(1-a^2) + \sqrt{(R(1-a^2) - Qb^2)^2 + 4b^2QR}}{2b^2}$ by the quadratic formula

$K = \frac{abS}{R + b^2S}$ by substitution

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